Assignment 2: String and Array Problem-Solving Toolkit

Problem statement: Implement a collection of reusable JavaScript functions to solve common programming problems.

Tasks

1. Write a function to reverse a string without using `reverse()`.
2. Write a function to check whether a string is a palindrome.
3. Write a function to count vowels and consonants in a string.
4. Write a function to find the second-largest distinct number in an array.
5. Write a function to remove duplicate elements from an array without using `Set`.
6. Write a function to find the sum of all even numbers in an array.
7. Write a function to find the frequency of a given element in an array.
8. Write a function to check whether two strings are anagrams.
9. Write a function to move all zeros to the end of an array while maintaining the order of non-zero elements.

Sample input and output

```
// Reverse a string
reverseString("hello");
// Output: "olleh"

// Check palindrome
isPalindrome("madam");
// Output: true

// Second-largest distinct number
secondLargest([10, 5, 20, 8, 20]);
// Output: 10

// Remove duplicates
removeDuplicates([1, 2, 2, 3, 1]);
// Output: [1, 2, 3]

// Move zeros
moveZeros([0, 1, 0, 3, 12]);
// Output: [1, 3, 12, 0, 0]
```

Bonus challenge: For each solution, explain the approach in 2–3 sentences and identify the time complexity where possible.